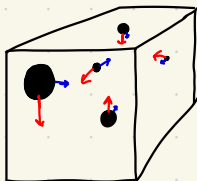


Goal: Plot an assortment of orbital bodies with randomly generated Velocity vectors in a bounded box. Go through each body and calculate the vector of gravitational force on it. Then add a visible red arrow for the velocity and a blue one for gravity.

e.g.



System Defaults

Unit of Mass = 1 Solar mass (SunMass)

Unit of distance = 1 Astronomical unit

Unit of time = 1 day

Unit of energy = $\frac{(\text{Solar mass}) \cdot (\text{AU})^2}{(\text{Day})^2}$

Gravitational Constant = 0.000295913120346 (See 1.1)

New Gravitational Constant 1.1

Current $G = 6.6743 \times 10^{-11} \frac{\text{m}^3}{\text{kg} \cdot \text{s}^2}$

New $G = 6.6743 \times 10^{-11} \frac{(\text{Solar mass}) (\text{day})^2}{(\text{AU})^3}$

- AU = 149,597,870,700 meters

- Solar Mass = 1.988416×10^{30} Kg

- Day = 86,400 seconds

New $G = 0.000295913120346$

Equation 1.2

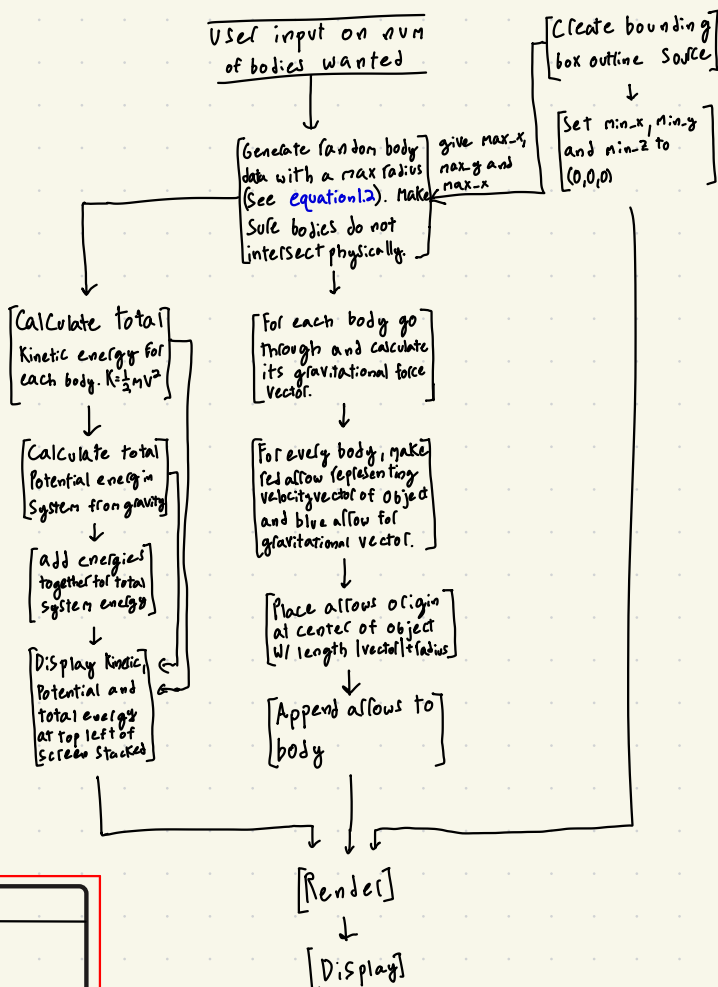
X # of bodies with variable radius r in a volume. Let q be a random ratio that determines max density of bodies wanted in system. The max variable size of r ($\text{max}(r)$) is determined by equation: $\text{Max}(r) = \frac{q}{2} \sqrt[3]{\frac{a}{x}}$

Custom Classes

Body
~Position Vector
~radius: Double
+ distance Squared (Body): Double
+ distanceTo (Body): Double
+ distanceTo (Body): Bool
+ isIn Bounds(): Bool

NewtonianBody
- Velocity Vector
- Gravity Vector
- mass: Double
+ GRAVITATIONAL_CONSTANT
+ setGravity(bodies[]): void
+ getMyKinetic(): Double
+ getMyPotential(): Double
+ getSystemKinetic(bodies[]): Double
+ getSystemPotential(bodies[]): Double

Functional Diagram



List of Objects

- NewtonianBody
- vtkGlyph3D
- vtkDoubleArray
- vtkMapper
- vtkPoints
- vtkActor
- vtkArrowSource
- vtkTextActor
- vtkSphereSource
- vtkRenderer
- vtkCubeSource
- vtkRenderWindow
- vtkPolyData
- vtkRenderWindowInteractor

Object Diagram

[# of bodies]



[NewtonianBody[]]
list of randomly generated
NewtonianBody objects
w/ max possible radius
following equation 1.1



[vtkDoubleArray]
Array of gravity
vector for each body

[vtkDoubleArray]
Array of velocity
vector for each body

[vtkDoubleArray]
Array of radius
for each body

[vtkPoints]
list of position
vector for
each body

[vtkArrowSource]
Velocity vector

[vtkArrowSource]
Gravity vector

[vtkSphereSource]
basis for each body

[vtkPolyData]
Stores the following:
Points: position array
Array: radius
Array: Velocity vector
Array: Gravity vector

[vtkTextActor]
Total kinetic
energy

[vtkTextActor]
Total system
energy

[vtkTextActor]
Total potential
energy

[vtkCubeSource]
Cube covering the
bounding box area

[vtkMapper]

[vtkActor]
Set representation
of property to
wire frame

[vtkGlyph3D]
Places velocity
arrows on screen
at their point

[vtkMapper]
maps vel. glyph

[vtkActor]

[vtkGlyph3D]
Places gravity
arrows on screen
at their point

[vtkMapper]
Maps grav. glyph

[vtkActor]

[vtkGlyph3D]
Places spheres at
their specified
coords and with
their radius

[vtkMapper]
Maps body glyph

[vtkActor]

[Renderer]



[vtkRenderWindowInteractor] → [Window]